



LeetCode #1768 — Merge Strings Alternately

Problem

Merge two strings by alternating their characters. If one string is longer, append the remaining characters to the end.

Approach

- **Pattern:** Two Pointers
- Traverse both strings simultaneously.
- Append one character from each string alternately.
- After one string is exhausted, append the remaining characters from the other string.

Algorithm

1. Initialize two pointers (`i`, `j`).
2. Alternate characters from both strings.
3. Append remaining characters from the longer string.
4. Return the merged string.

Complexity

- **Time Complexity:** $O(n + m)$
- **Space Complexity:** $O(n + m)$

Key Learning

- Two Pointers technique
- Efficient string construction in Python using `"".join()`
- Handling unequal-length inputs
- Writing clean and readable solutions

Language: Python 🐍

★ Consistently solving LeetCode problems to strengthen Data Structures & Algorithms and interview preparation.